

Seat No: .....

Enrollment No: .....

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**B.Tech, Winter 2025-26 Examination**

Semester: VI

Subject Code: 303105341

Subject Name: Cyber Security

Date: 18-11-2025

Time: 10.30 am to 1.00 pm

Total Marks: 60

**Instructions:**

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

**SECTION - A**

| <b>Q.1</b> | Answer the following questions.                                                                                                                                                                                                    | <b>Marks</b> | <b>CO</b>   | <b>BT</b> |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|-----------|
|            | A. a) Name the attack method that guesses passwords by trying multiple combinations.<br>b) Explain how this attack works in practice.<br>c) Mention one technique to prevent or mitigate this type of attack.                      | 06           | CO1,<br>CO4 | BT2       |
|            | B. a) Mention two steps involved in scanning a cloud-based network infrastructure.<br>b) Explain one unique challenge when scanning cloud environments.<br>c) Describe one security measure to overcome cloud scanning challenges. | 06           | CO1         | BT2       |
| <b>Q.2</b> | A. Define the "default deny" policy in firewalls.                                                                                                                                                                                  | 04           | CO1,<br>CO5 | BT4       |
|            | B. How does port forwarding work? Can you explain the steps simply?                                                                                                                                                                | 05           | CO2,<br>CO5 | BT1       |
|            | <b>OR</b>                                                                                                                                                                                                                          |              |             |           |
|            | B. How do security policies help in protecting organizational data?                                                                                                                                                                | 05           | CO2,<br>CO3 | BT1       |
| <b>Q.3</b> | A. Explain the significance of what is steganography, and how is it used in cybersecurity? in a practical scenario.                                                                                                                | 04           | CO3,<br>CO6 | BT2       |
|            | B. What is the role of an incident response plan within cybersecurity?                                                                                                                                                             | 05           | CO6         | BT4       |
|            | <b>OR</b>                                                                                                                                                                                                                          |              |             |           |
|            | B. Why is describe how network address translation (nat) is used in firewalls. important in the field of cybersecurity?                                                                                                            | 05           | CO3,<br>CO4 | BT2       |

**SECTION - B**

| <b>Q.1</b> | Answer the following questions.                                                                                                                                                                             | <b>Marks</b> | <b>CO</b> | <b>BT</b> |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|-----------|
|            | A. a) Mention the first phase in executing a computer-related offense.<br>b) Describe the second phase of the offense execution process.<br>c) Explain the third phase and its impact on the target system. | 06           | CO1       | BT2       |
|            | B. a) What is a Management Information System (MIS)?<br>b) Mention two ways MIS helps organizations gain a competitive advantage.                                                                           | 06           | CO1       | BT5       |

|            |                                                                                                               |    |             |     |
|------------|---------------------------------------------------------------------------------------------------------------|----|-------------|-----|
|            | c) Provide one example of how MIS enhances decision-making for a competitive edge.                            |    |             |     |
| <b>Q.2</b> | A. What is the significance of open ports in a network scan, and how can these contribute to vulnerabilities? | 04 | CO3         | BT2 |
|            | B. Explain the concept of network reconnaissance and how it aids in security testing.                         | 05 | CO2,<br>CO3 | BT3 |
|            | <b>OR</b>                                                                                                     |    |             |     |
|            | B. List and briefly explain common vulnerabilities detected during vulnerability scanning.                    | 05 | CO2,<br>CO3 | BT5 |
| <b>Q.3</b> | A. What does hacking mean, and what are its different types?                                                  | 04 | CO4         | BT6 |
|            | B. What is social engineering in the context of cybercrime? Identify three social engineering methods.        | 05 | CO4         | BT3 |
|            | <b>OR</b>                                                                                                     |    |             |     |
|            | B. Differentiate between data, information, and knowledge.                                                    | 05 | CO1,<br>CO3 | BT3 |
|            |                                                                                                               |    |             |     |